

YOUSSEF YASSER ELSAYED HAMED

Associate AI & Software Engineer

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🐙 github.com/itz-youssef 🌐 [Portfolio](#) 📄 [Kaggle](#) 📍 Giza, Egypt 12513 (Open to Relocation & Remote)

PROFILE

Innovative junior Software and AI Engineer dedicated to architecting robust backend infrastructures and state-of-the-art artificial intelligence systems. Highly skilled in leveraging applied deep learning, advanced computer vision, and autonomous agentic workflows to solve complex computational challenges. Committed to applying rigorous software engineering principles to deploy secure, production-grade intelligent solutions.

EDUCATION

Cairo University

Bachelor's in Computer Science and Artificial Intelligence

Giza, Egypt

Sep. 2023 – Sep. 2027

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Signals & Systems, Generative Adversarial Networks (GANs), Supervised & Unsupervised Learning, Robotics & Autonomous Multiagent Systems, Probability & Statistics

EXPERIENCE

AI Trainee Huawei ICT Academy Program | [Certificate](#)

Huawei ICT Academy-Egypt

Sep. 2026 – Oct. 2026

Remote, Cairo, Egypt

- Enrolled in the Huawei Certified ICT Associate – Artificial Intelligence (HCIA-AI) training program, an intensive, industry-recognized curriculum covering the full AI development stack.

Computer Vision Engineer Intern | [Certificate](#)

National Telecommunication Institute (NTI)

Jul. 2026 – Aug. 2026

Hybrid, Cairo, Egypt

- Developed the capstone project, **Driver Fatigue & Attention Monitoring System** 🌀, to detect prolonged eye closure and yawning in real-time using **MediaPipe Face Mesh** and a custom **MobileNetV2 CNN**.
- Engineered a live **Flask/SQLite** dashboard with temporal tracking to filter false positives from normal blinking, featuring multi-threaded audio alerts and a driving-time tracking interface.

Artificial Intelligence Intern | [Certificate](#)

Aitronix

Jun. 2026 – Aug. 2026

On-site, New Cairo, Egypt

- Worked on production-level AI systems spanning computer vision, NLP, and LLM integration, leveraging **FastAPI** and data analysis techniques.
- Engineered an automated customer support pipeline using **n8n**, deploying a **WhatsApp** bot for a restaurant client to instantly process user inquiries and trigger immediate, context-aware replies.

AI Training Evaluator (Freelance)

Outlier

Jun. 2026 – Aug. 2026

Remote, Cairo, Egypt

- Evaluate and rate conversational AI model outputs for **RLHF** training pipelines across dimensions including naturalness, engagement, utility, and audio quality.

Computer Vision Intern

Aitronix

Apr. 2026 – Jun. 2026

On-site, New Cairo, Egypt

- Specialized in building and optimizing computer vision pipelines for real-world deployment, integrating **Vision Science** and **Retrieval-Augmented Generation (RAG)**.
- Engineered a real-time **Behavioral AI Theft Detection** system with multi-object tracking and Re-ID, using **YOLOv11 + ByteTrack + LSTM** to classify suspicious human-object interactions.
- Deployed **YOLOv8s** object detection models directly onto the enterprise's live IP camera network via RTSP, achieving real-time inference and continuous spatial monitoring on live streams.

COMPETITIONS

Competitive Programmer | [Certificate](#)

Egyptian Collegiate Programming Contest (ECPC)

July 2024

Alexandria, Egypt

- Engineered optimized algorithmic solutions in **C++** under strict time constraints, applying advanced data structures to tackle complex computational problems.
- Collaborated in a high-pressure team environment to prioritize problem selection, rapidly debug logical errors, and maximize competitive performance.

PROJECTS

YAI: Video Safety Analyzer [GitHub] [Live Demo] <i>PyTorch, FastAPI, CLIP, ByteTrack, Redis</i>	May 2026
• End-to-End CV Pipeline: Engineered a real-time video safety analysis system with an async FastAPI backend, processing footage through a 5-stage pipeline: detection, open-vocabulary classification, tracking, anomaly scoring, and depth estimation. • Multi-Model Inference: Integrated YOLOv10x, ByteTrack with Kalman filtering for persistent tracking, AnomalyTransformer for behavioral threat scoring across 6 categories, and Depth Anything v2 for monocular depth.	
APEX: Adaptive Pose & Action Intelligence System [GitHub] <i>PyTorch, LSTM, SQLite</i>	June 2026
• Action Recognition: Developed a real-time multi-modal AI system for intelligent human action recognition using YOLOv8n-pose for high-speed 17-keypoint skeleton tracking and posture analysis (8 critical joint angles). • Temporal Sequence Modeling: Designed a hybrid pipeline combining rule-based motion analysis with an LSTM-based network processing 30-frame continuous sequences to classify complex actions and posture transitions.	
Pie: AI-Powered Marketing Kit Generator [Live Demo] <i>Groq, VLMs, RAG, FastAPI, React</i>	August 2026
• Generative AI & RAG: Built a full-stack AI tool that transforms product screenshots into comprehensive marketing kits (captions, hooks, influencer briefs) in under 60 seconds using Vision-Language Models (VLMs). • High-Speed Inference: Powered by Groq for ultra-fast text generation and a custom Retrieval-Augmented Generation (RAG) pipeline that continuously scales and improves from user feedback.	
Behavioral AI Theft Detection System [GitHub] <i>PyTorch, YOLOv11x, SAHI, ByteTrack, Supervision</i>	April 2026
• Robust Detection & Tracking: Engineered a pipeline using YOLOv11x and SAHI (1000×1000 tiling) to maximize small-object recall, with ByteTrack and a ResNet18-based ReID manager for multi-target tracking. • Behavioral Classification: Extracted 17 dynamic spatial features (grab motion, joint angles) to train a custom PyTorch LSTM network with an attention mechanism for classifying suspicious temporal movements.	
Smart Agriculture Crop Disease Detection [Kaggle] <i>ViT, Grad-CAM, Gradio</i>	Feb - May 2026
• Automated Diagnosis & Kaggle Bronze Medal: Built a disease classification system trained on 54,000+ images. Benchmarked 14 distinct models (ViT, MobileNetV2, LSTMs, Attention-CNNs). Awarded a Kaggle Bronze Medal for the comprehensive model evaluation notebook.	

CERTIFICATES & AWARDS

Microsoft Applied Skills: Secure AI Solutions, Cloud Security & Entra ID <i>Microsoft</i>	Sep. 2026
Huawei Certified ICT Associate (HCIA-AI V4.0) <i>Huawei ICT Academy (ID: ICT20261004014238)</i>	Oct. 2026
Google AI Essentials <i>Google – Coursera</i>	Sep. 2026
Claude Code in Action (Agentic Workflows & LLMs) <i>Anthropic</i>	June 2026
Getting Started with Deep Learning <i>NVIDIA Deep Learning Institute</i>	Dec. 2025
Master OpenCV Fundamentals for Real-Time CV <i>Coursera – EDUCBA</i>	May 2026
Deep Learning Specialization <i>DeepLearning.AI (Andrew Ng) – Coursera</i>	Feb. 2026

TECHNICAL SKILLS

Programming & Core: Python, C++, CUDA, Java, JavaScript, SQL, C#

Generative AI & LLMs: Vision-Language Models (VLMs), Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Agentic AI workflows, Diffusion Models, Zero-Shot Learning

Deep Computer Vision: Multi-Object Tracking, Pose/State Estimation, Instance & Semantic Segmentation, Monocular Depth, Open-Vocabulary Classification, Small-Object Detection (SAHI), Anomaly Detection

MLOps & Cloud Infrastructure: AWS (SageMaker), Google Cloud (Vertex AI, BigQuery), Azure, Kubernetes (K8s), Docker, CI/CD (GitHub Actions), MLflow, TensorRT, ONNX, Edge Inference

Frameworks & Ecosystems: PyTorch, Hugging Face, LangChain, LlamaIndex, TensorFlow/Keras, OpenCV, YOLO (v8-v11-v13-v26), CLIP, ByteTrack, MediaPipe, Pandas, Scikit-Learn

Data Engineering & Backend: FastAPI, REST APIs, WebSockets, Vector Databases (ChromaDB, Pinecone), PostgreSQL, Supabase, End-to-End Pipeline Architecture

LANGUAGES

Arabic: Native

English: Professional Working Proficiency (B2+ / Duolingo English Test: 130)